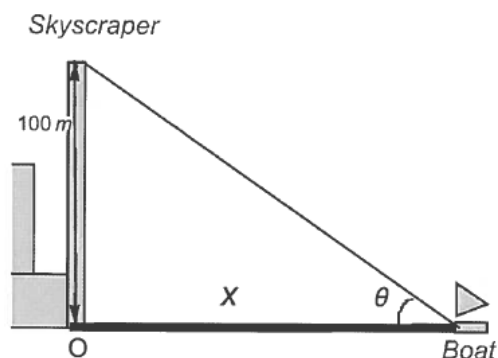


Problem 7

Problem 7

Problem 1 A boat sails directly toward a 100-meter skyscraper that stands on the edge of a harbor. The angular size θ of the building is the angle formed by lines from the top and bottom of the building to the observer on the boat (see figure below).



- (a) Express the angle θ as the function of x , the distance of the boat from the building.

Explanation. $\tan \theta = \frac{100}{x} \implies \theta = \tan^{-1} \left(\frac{100}{x} \right)$

- (b) Find the angular size, θ , when the boat is $x = 100\sqrt{3}m$ from the building.

Explanation. $\theta = \tan^{-1} \left(\frac{100}{100\sqrt{3}} \right) = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6}$